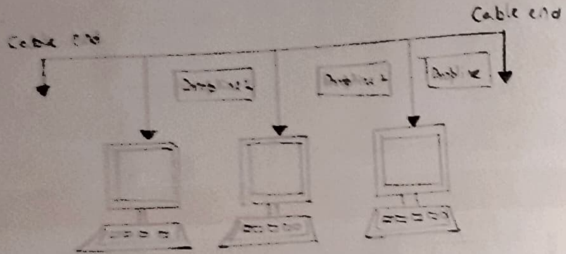


Bus Topology

Aim:- Study of different types of network topologies.

Types of Network Topology

Network topology is the schematic description of a network arrangement, connecting various nodes (sender & receiver) through lines of connection.

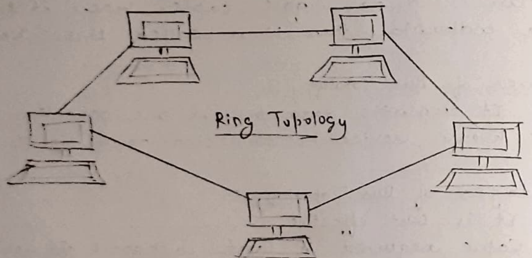
* Bus Topology Bus topology is a network type in which every computer & network device is connected to single cable. When it has exactly two endpoints, then it is called Linear Bus Topology.

Features of Bus Topology:-

1. It transmits data only in one direction.
2. Every device is connected to a single cable.

Advantages of Bus Topology:-

1. It is cost effective
2. Cable required is least compared to other network topology.
3. Used in small networks.
4. It is easy to understand.
5. Easy to expand joining two cables together.



Disadvantages of Bus Topology:-

1. Cables fails then whole network fails.
2. If network traffic is heavy or nodes are more the performance of the network decreases.
3. Cable has a limited length.
4. It is a slower then the ring topology.

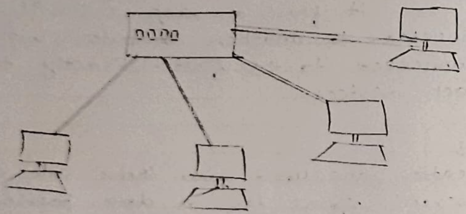
* **Ring Topology:-** It is called ring topology because it forms a ring such as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbors for each devices.

Features

1.) Repeaters are used when there are many nodes to prevent signal loss as data passes through multiple nodes.
2.) Data transmission is usually unidirectional in a ring topology.
3.) Dual Ring topology uses two rings where data flows in opposite directions for backup & reliability.
4.) Sequential data transfer - data travels node by node (bit by bit) until it reaches the destination.

Advantages

1.) Transmission network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
2.) Cheap to install & expand.



Star Topology

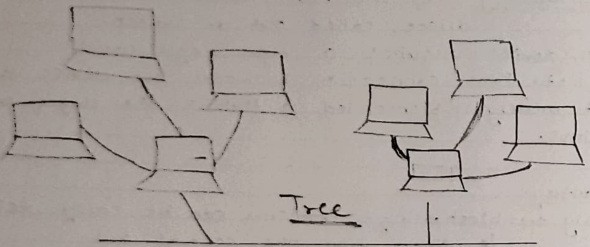
Disadvantage

1. Troubleshooting in difficult in ring topology.
2. Adding or deleting the computers disturbs the network activity.
3. Failure of one computer ~~distributes~~ disturbs the whole network.

Star Topology:- All computers / devices connect to a central device called hub or switch.
Each device requires a ~~sign~~ single cable.
Point-to-Point connection between the device & hub.
Most widely implemented & Hub is the single point of Failure.

Advantage

1. Easy troubleshooting - Problems can be easily detected & fixed from the central hub.
2. Better network control - Network management & modification are easy.
3. Limited failure - failure of one cable or device does not affect the whole network.
4. Cost-effective technology - Uses inexpensive networking equipment.
5. Easy to expand - New devices can be added easily to the hub or switch.
6. High data speed - Supports high bandwidth (around 100 mbps ethernet)



Tree

Disadvantage

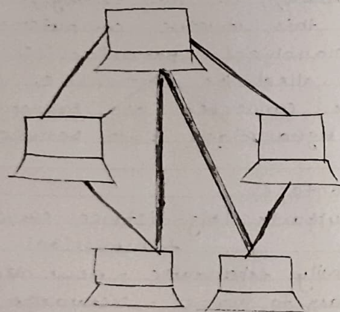
1. Central point of failure - if the central Hub or switch fails, the entire network stops working.
2. More cable required - large network require more cabling, which can make installation & routing difficult.

* Tree Topology:-

Tree topology combines the features of bus Topology & star topology.
In this structure, computers are connected in a hierarchical (parent-child) manner. The Top node is called the root node. & all the other nodes are connected as branches. Data travels through a ~~signa~~ single path between nodes.

Advantages

1. Supports long-distance communication (broadband transmission)
2. Easily expandable - new devices can be added easily.
3. Easy to manage - Networks is devided into smaller star segements.
4. Easy error detection - faults can be identified quickly.
5. Limited failure - failure of one node does not affect the whole network.
6. Point-to-point wiring is used for each segement.



Mesh Topology

Disadvantages

1. Difficult troubleshooting - It is hard to find faults in the networks.
2. High-cost - Devices used for broadband transmission are expensive.
3. Main cable dependency - Failure of the main bus Cable can stop the entire network.
4. Reconfiguration is difficult - Adding new devices may require network changes.

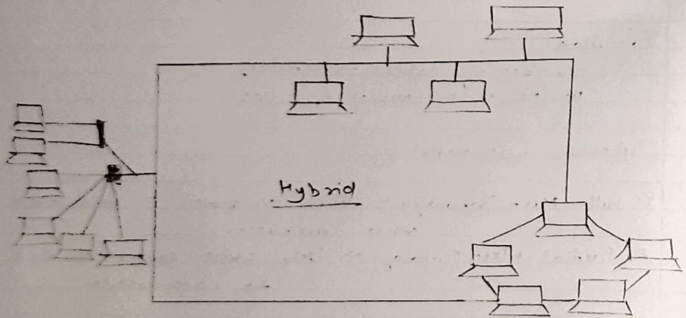
* Mesh Topology Mesh topology is a network structure where each computer is connected to other computers through multiple paths. It doesn't use a central device like a hub or switch. The Internet is a common example of mesh topology.

Formulas.

Number of cables => $n(n-1)/2$
where n = number of nodes.

Types of mesh topology

1. Full Mesh Topology -> Every computer is connected to all other computer.
2. Partial Mesh Topology -> Only some computers are connected to each-other.



Advantage

1. Reliable - failure of one link does not affect the network.
2. Fast communication - Multiple paths allow quick data transfer.
3. Easy reconfiguration - New devices can be added without disturbing others.

Disadvantage

1. High-cost => Requires many cables & networking devices
2. Difficult management => large networks are hard to maintain.
3. Lower efficiency => Many redundant connections reduce efficiency.

* **Hybrid Topology** :- It is formed by combining two or more different network topologies (such as star, bus or ring). It connects different network structures to improve performance & flexibility.

Advantage

1. Reliable - Failure in one part of the network does not affect the entire network.
2. Scalable - Network size can be expended easily by adding new devices.
3. Flexible - can be designed according to organisational needs
4. Effective - Combines strengths of different topologies & reduces their weakness

Disadvantage

1. complex design - Network architecture is difficult to design.
2. Costly Hubs - Specialized hubs & devices are expensive.
3. High infrastructure cost - Requires more cabling & networking devices.